
Venkat Neelraj Nitta

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Summary

Software and AI developer with over 2 years of experience specializing in Deep Learning, Data Science, and Generative AI (Python, PyTorch, LLMs, Computer Vision). Experienced **AI and NLP Specialist** skilled at building end-to-end pipelines for both text and image data. Built an **AI-powered document preprocessing** system using computer vision to ensure flawless quality control, which sped up **invoice processing** by **90%** and **cut costs** by **60%**. Led a team to optimize a **cheque processing model** that achieves perfect **key value extraction** and slashes processing time to just **7 seconds per cheque**. Combines analytical insight with practical problem-solving to deliver impactful AI-driven solutions.

Education

Northeastern University, Boston, MA **Sep 2024 – Expected Apr 2026**

Master of Professional Studies in Applied Machine Intelligence, STEM GPA: 4.0

Related courses: Fundamentals of Artificial Intelligence, Applications of AI, AI System Technologies (ML Ops), Healthcare Information Processing, Healthcare/Pharmaceutical Data and Applications.

Mahindra Ecole Centrale, Hyderabad, India **Jul 2022**

Bachelor of Engineering in Electrical Electronics Engineering

Related courses: Signals and Systems, Data Structures, Numerical Methods, Statistics, Digital Signal Processing, Machine Learning, Natural Language Processing, Advance Data Analytics, Big Data Computing.

Technical Knowledge

Programming and Development	Python, SQL, Pytorch, Django, FastAPI, Docker, Linux (Ubuntu).
Deep Learning and Generative AI	Large Language Models (LLMs), Transformers, NER, Transfer Learning, Fine-Tuning, Quantization, LoRA, QLoRA, Inference Optimization, OpenCV, OCR, Vector Databases.
Data Engineering and Analytics	Data Processing, Feature Engineering, PySpark, MySQL, Hadoop, Snowflake, PostgreSQL, Tableau.

Work Experience

Software/AI Developer, Cognida.ai – Hyderabad, India **Jul 2022 – Aug 2024**

- Provided **AI and NLP Specialist** expertise by developing and fine-tuning an **enterprise-grade conversational AI chatbot** leveraging advanced Natural Language Processing and Machine Learning techniques, reducing query resolution time from **10 days to minutes** and elevating customer satisfaction by **78%**
- Created an **AI-powered document preprocessing system** using OCR, computer vision, and neural networks to classify image quality and extract textual data, resulting in a **100% compliance rate**, **90% faster invoice processing**, and **60% cost savings** through streamlined workflows and minimized manual errors.
- **Directed the AI team** in training and optimizing a cheque processing pipeline using **Named Entity Recognition (NER)** and **object detection models**, achieving **100% key data extraction accuracy**, reducing processing time per cheque to just **7 seconds**, and boosting overall document workflow efficiency by **40%**.

Software/AI Developer Intern, Cognida.ai – Hyderabad, India **Feb 2022 – Jun 2022**

- Developed and optimized an NLP-based **language translation and grammar correction model**, using **Named Entity Recognition (NER)** techniques to ensure context-aware, accurate corrections with real-time performance.
- Designed an interactive **feedback system** that allows users to **validate, override, and refine corrections**.

Hackathons

InnovAItE, Northeastern: Smart Cart App Feb 2025

Github: [Smart-Cart](#) **Tools:** Django, PostgreSQL, React, Google Gemini API, Hugging Face

- Led the development of SmartCart, an app focused on providing competitive pricing by **automatically identifying the lowest vendor prices** and reducing grocery shopping time.
- **Directed a team to automate vendor comparisons**, optimizing the shopping process and improving user efficiency while supporting vendors with better pricing strategies and inventory management.

Cognida Cognida Competition, Cognida.ai: Object Detection Web Application Feb 2022

- **Won the competition** and secured an internship offer to achieve the **highest precision among all participants**.
- Developed an object detection web application using **PyTorch**, overcoming the challenge of limited available data by using **web scraping** techniques to gather relevant data and optimize model performance.

Projects

Electric Vehicles in WA: Price, Type, and Range Feb 2025

Tableau Dashboard: [Tableau Public Link](#)

- Designed a data-driven **dashboard** that helps customers compare EV models **based on price, range, and type before making a purchase**.
- Provided **visual insights** on **affordability, battery type, and model popularity**, allowing users to assess their best options

CIFAR-10 Image Classification Web App Dec 2024

GitHub: github.com/Subhash-269/cifar-image-cls-app

Live App: cifar-image-classifier.streamlit.app

Technologies: Python, PyTorch, FastAPI, Streamlit

- **Trained** and **fine-tuned** a ResNet-18 model, optimizing it for accurate classification across 10 categories and enhancing deployment readiness.
- Implemented **model quantization** and integrated an **SQLite** database, improving inference speed while enabling **storage, tracking, and correction** of user-uploaded images and predictions.

Brain Tumor Detection Feb 2022 - Mar 2022

Technologies: Deep Neural Network, Object Detection, Transfer Learning

- Developed a **deep learning model** for brain tumor detection, initially training on one tumor type using a dataset of 6,000 **medical images**
- Expanded its scope to classify additional tumor types using **transfer learning**, achieving over **85%** precision while improving efficiency.

Certifications

- **MLOps; Machine Learning in Production, DeepLearning.AI** Apr 2025 - Present
- **Generative AI with Large Language Models, DeepLearning.AI** May 2024
- **Applying AI Technologies to the Workplace, Northeastern University** Sep 2024
- **Foundations of Modern Machine Learning, IIIT-Hyderabad** Jul 2022
- **Intro to Machine Learning Course, Kaggle** 2021
- **Intermediate Machine Learning Course, Kaggle** 2021